

Assignment-based Subjective Questions

1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable? (3 marks)

Categorical variables in the dataset:

- season
- yr
- mnth
- holiday
- weekday
- workingday
- weathersit

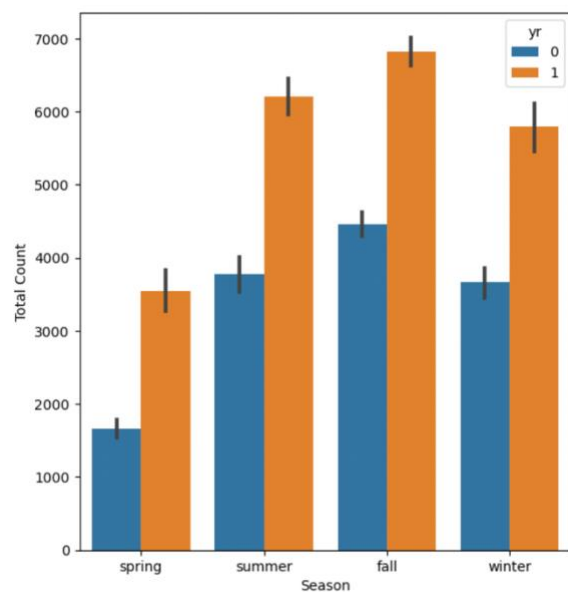
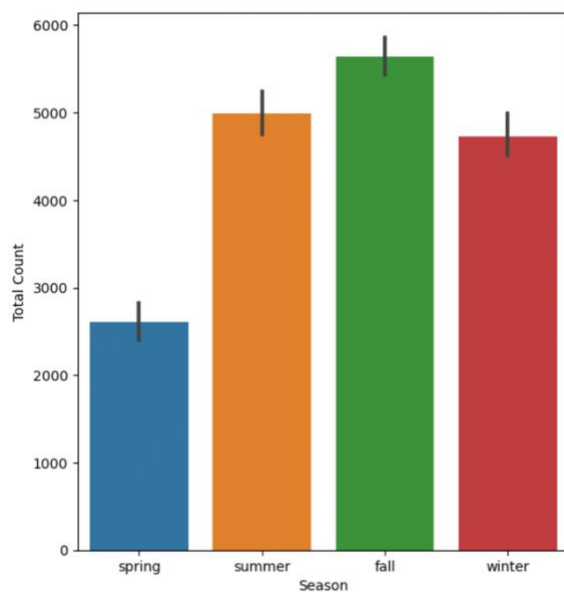
Season:

Unique values included spring, summer, fall, winter

Booking count reached maximum in fall season

Booking count was minimum in spring season

Counts in all seasons increased by a good margin from 2018 to 2019

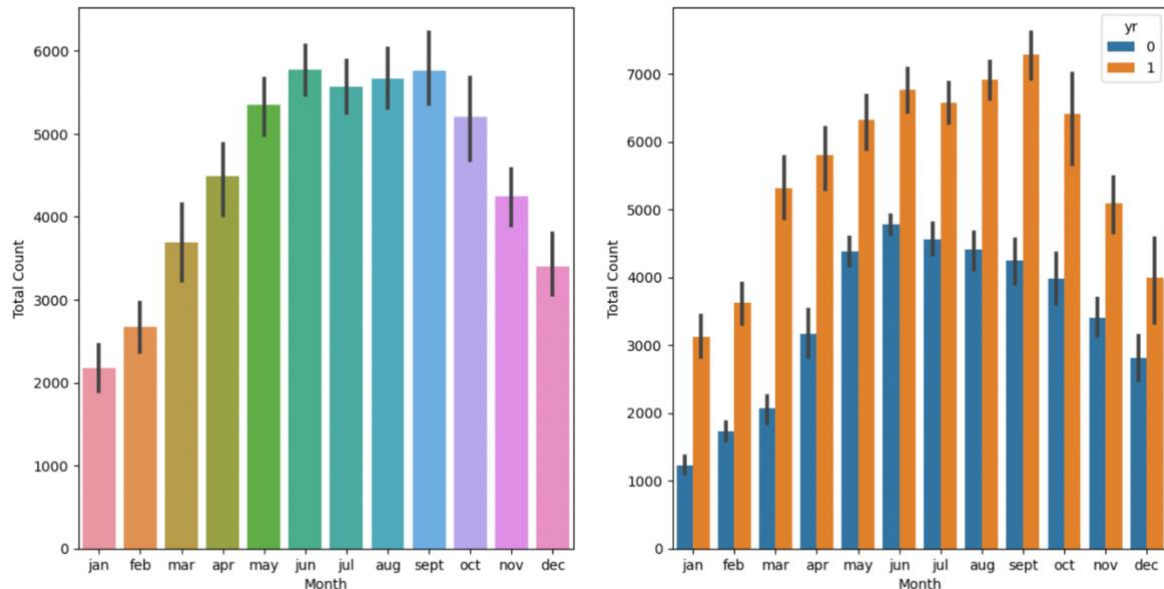


Month:

Unique values included jan, feb, mar, apr, may, jun, jul, aug, sept, oct, nov, dec

Bookings kept increasing month over month, reached peak around jun – sept, then started decreasing; roughly like a bell curve

Counts in all months increased by a good margin from 2018 to 2019

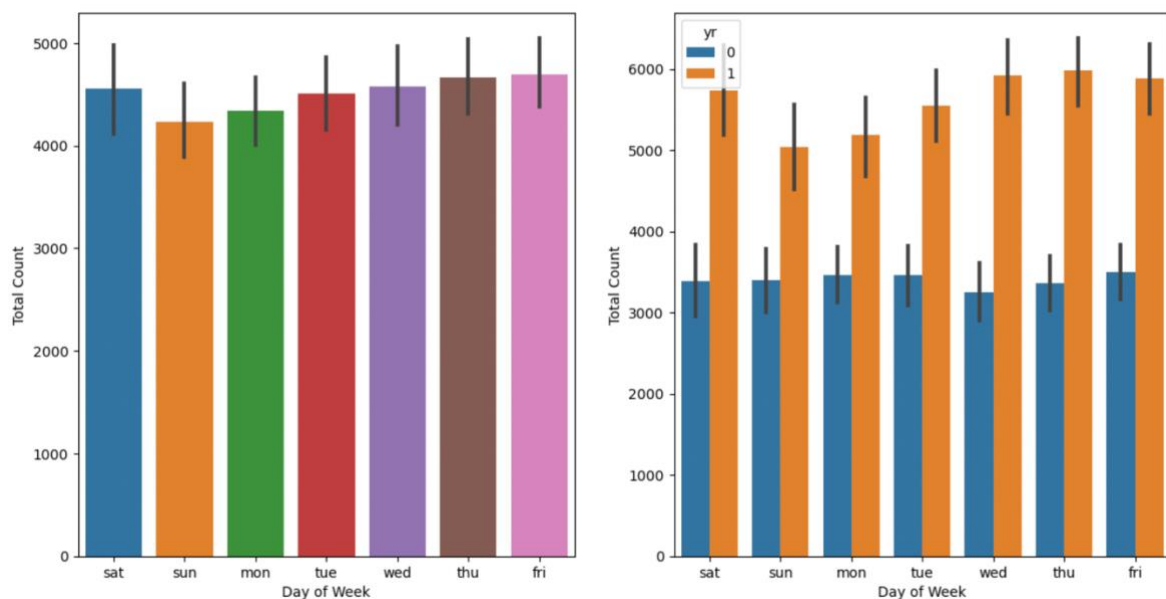


Weekday:

Unique values included sat, sun, mon, tue, wed, thurs, fri

Maximum bookings happened on Saturday, the amount dipped on Sunday, and then kept slightly increasing over the days

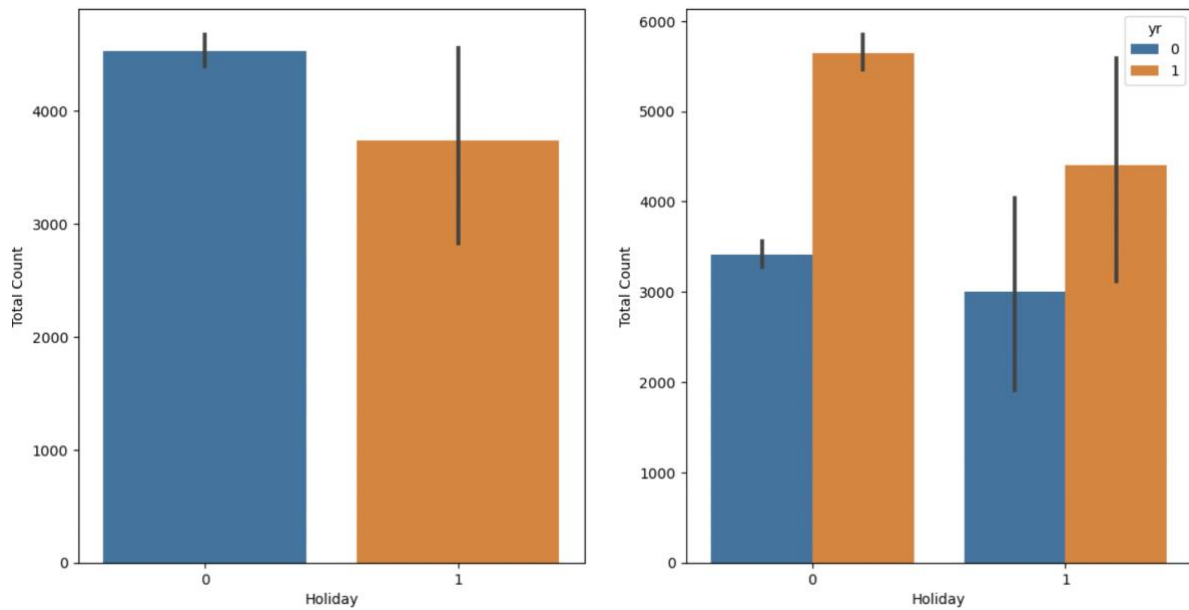
Counts in all days increased by a good margin from 2018 to 2019



Holiday:

Unique values included 0 or 1

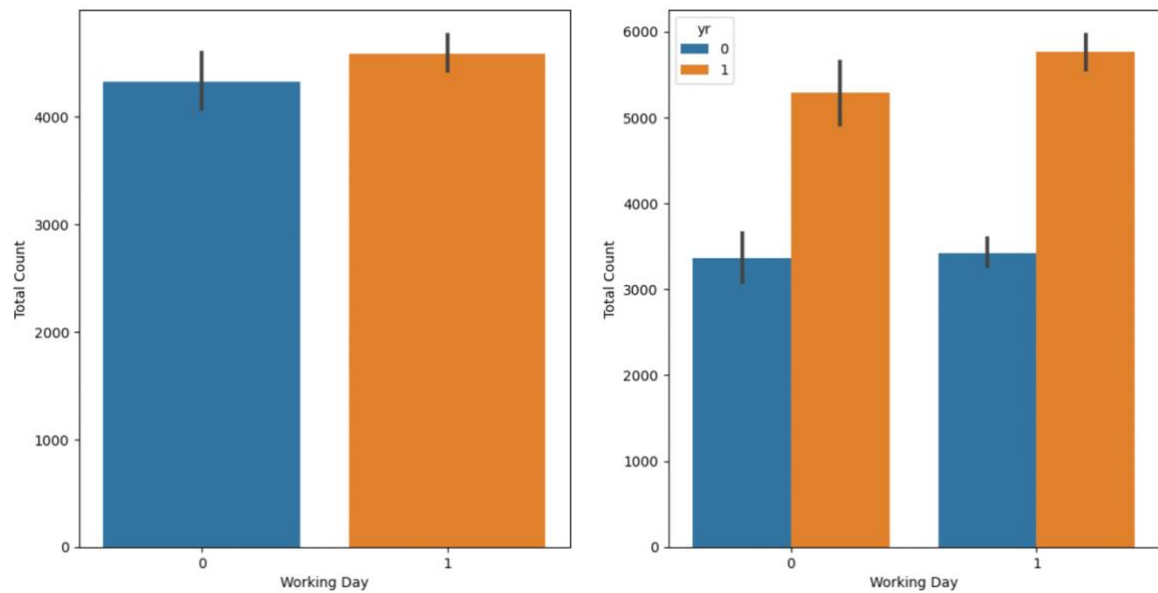
Booking demand seemed to decrease on holidays



Working day:

Unique values included 0 or 1

Booking demand seemed to increase on working days

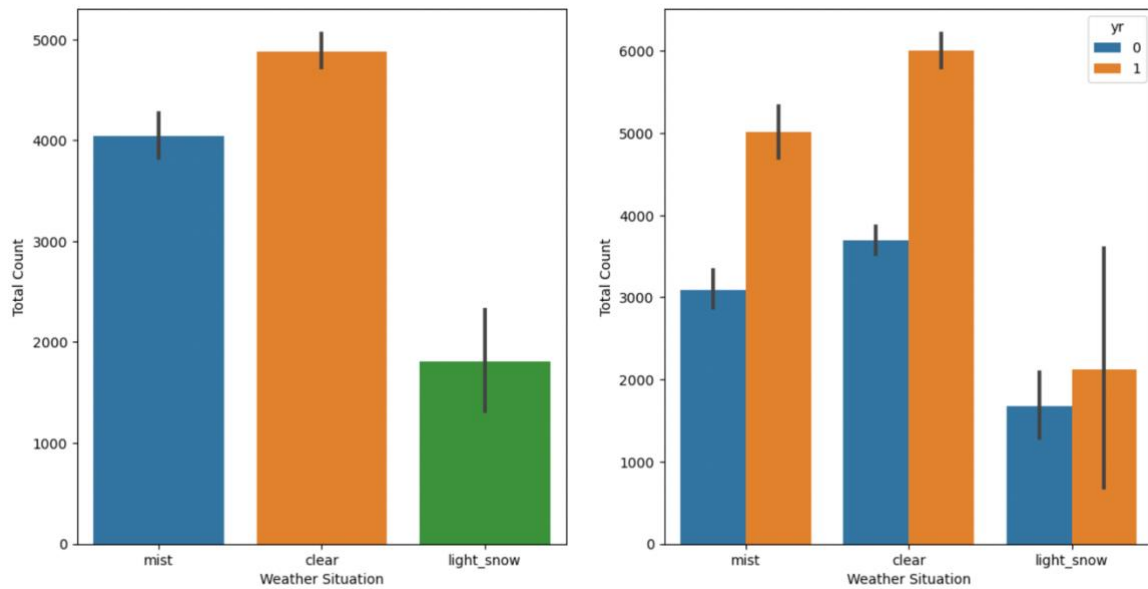


Weather situation:

Unique values included mist, clear, light_snow, heavy_rain

As can be guessed, booking demands were maximum in clear weather. Demands decreased on mist, even lesser in light_snow and practically none in heavy_rain

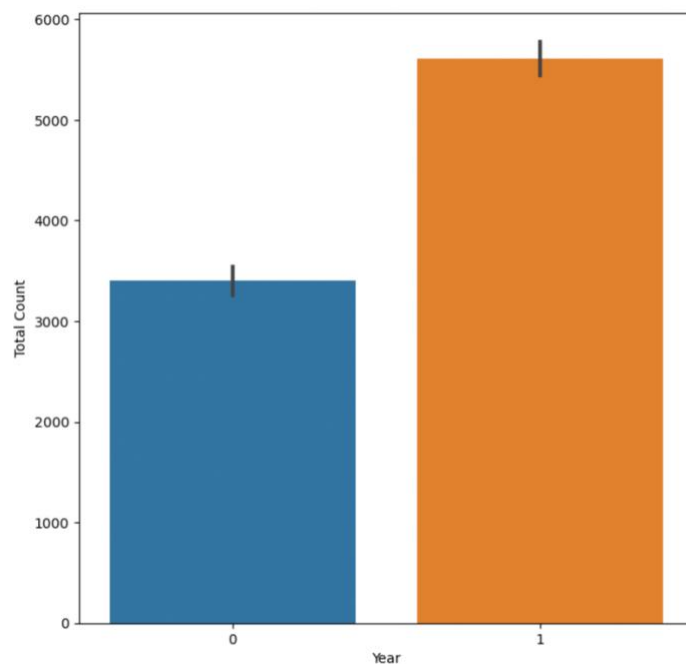
Counts in all weather situations increased by a good margin from 2018 to 2019



Year:

Booking demand increased by a very big margin from 2018 to 2019

Indicates good year over year progress



2. Why is it important to use **drop_first=True** during dummy variable creation? (2 mark)

Let's understand this with an example. Suppose we have a column which tells us about the state of furniture in houses up for sale. It can take the following possible values:

furnish_status
semi_furnished
furnished
unfurnished

If we create dummy variables out of this without **drop_first=True**, it'll create the following:

semi_furnished	furnished	unfurnished
1	0	0
0	1	0
0	0	1

So the encoding looks like this:

semi_furnished = 100

furnished = 010

unfurnished = 001

We could've easily done this using two variables with the following encoding:

semi_furnished	furnished
1	0
0	1
0	0

semi_furnished = 100

furnished = 010

unfurnished = 000

Hence, **drop_first=True** is important because:

1. It helps reduce the number of redundant variables added to the dataset
2. If we add a variable that isn't contributing much to the dataset and is dependent on other variables, we're at the risk of introducing multicollinearity to the model

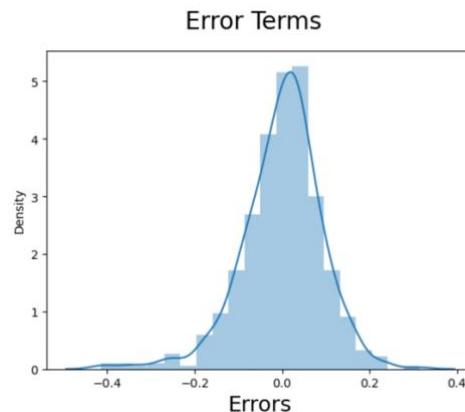
3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable? (1 mark)

Based on the pair plots and correlation matrix, **temp variable (temperature)** has the highest correlation with the target variable

4. How did you validate the assumptions of Linear Regression after building the model on the training set? (3 marks)

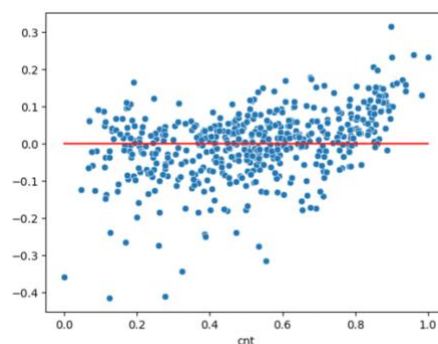
Assumptions of linear regression:

Error terms are normally distributed: We initially divide the data into training data and test data. Error terms are calculated by subtracting training data independent variable from the independent variable predicted by the model on training data. We can then plot the distribution of error terms and check if it is normal distribution.



Error terms are independent of each other: When building and re-building the model, we focus on reducing the variance inflation factor (VIF) to less than 5 for all variables. Low VIF means low multicollinearity which implies variables are not highly correlated with each other.

Error terms do not form patterns (homoscedasticity): The variance of error terms should be consistent across all levels of independent variables. We can check this by plotting a scatterplot of error terms versus the actual independent variable of training data and make sure they are not forming any type of visible patterns.



5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes? (2 marks)

- temp (**temperature**)
- yr (**year**)
- season_winter (**winter**)

General Subjective Questions

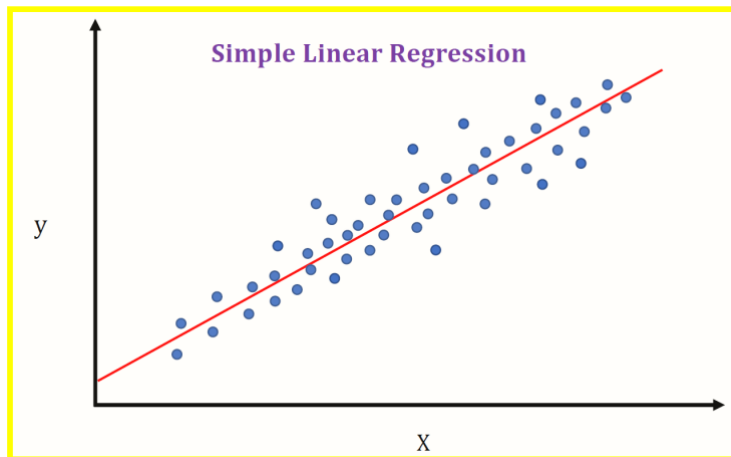
1. Explain the linear regression algorithm in detail. (4 marks)

Regression is a supervised machine learning algorithm that uses predictive analytics and is mostly used when the output variable that needs to be predicted is a numeric / continuous variable like sales prediction or marks of a student.

Linear regression predicts the relationship between two or more variables by assuming a linear relationship between the dependent and independent variables.

Simple Linear Regression:

When the dependent variable is predicted using only one independent/predictor variable, it is called simple linear regression. Simple linear regression revolves around finding the slope and intercept of the best-fit line on a plot where dependent variable is on y-axis and independent variable is on x-axis.



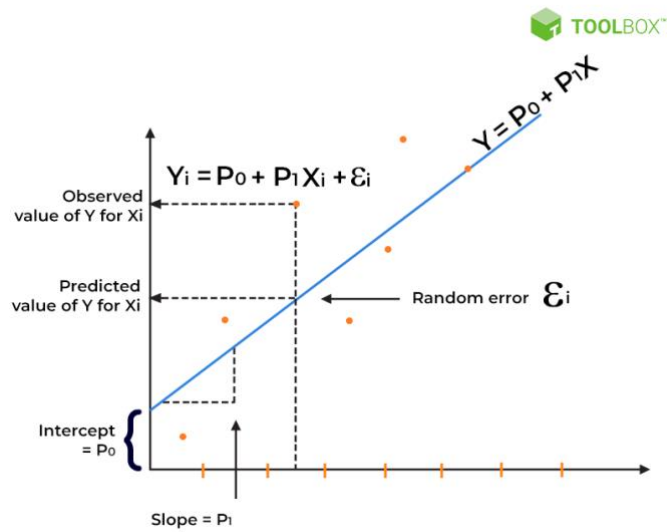
$Y = mx + c$ where m = slope and c = gradient (y intercept when $x = 0$)

The best-fit line is identified by minimizing the residual sum of squares (RSS). This is usually done through a cost function that optimizes the value of m and c . In turn, gradient descent is one of the main optimization algorithms which is used to optimize this cost function in order to reach the best possible solution with least RSS.

Assumptions of Simple Linear Regression:

- There is a linear relationship between dependent and independent variables.
- Error terms follow a normal distribution with mean = 0
- Error terms are independent of each other.
- Error terms have constant variance.

Model building for simple linear regression also starts with a null hypothesis that the slope for best-fit line is not significant, i. e, $m = 0$. The model is considered a good fit when we are eventually able to reject the null hypothesis by analysing the p-value of dependent variable and making sure it is almost equal to zero.

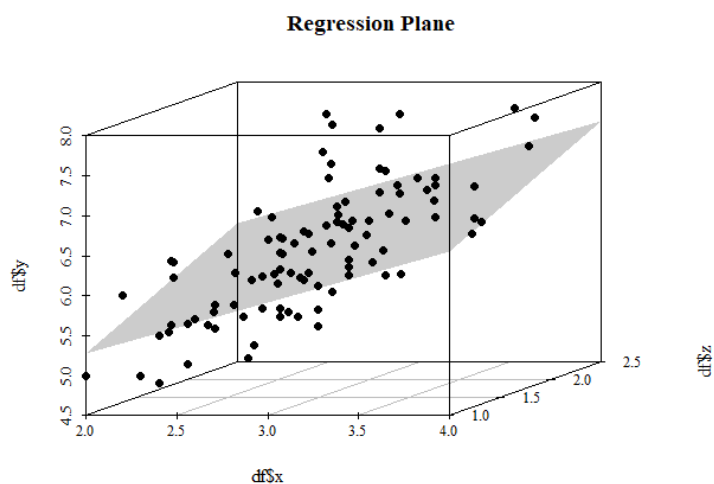


Multiple Linear Regression:

When the dependent variable is predicted using more than one independent/predictor variables, it is called multiple linear regression.

$$Y = c + m_1x_1 + m_2x_2 + \dots + m_nx_n$$

It is an extension to simple linear regression. Instead of finding the best-fit line, we find the best fit plane.



Additional considerations for Multiple Linear Regression:

In addition to the assumptions of simple linear regression, there are some considerations followed for multiple linear regression.

- Adding more isn't always helpful.
- Adding many variables may cause model to overfit and become complex.
- Feature selection is an important aspect.
- Risk of multicollinearity is always there.

Having related predictor variables in a dataset is called multicollinearity. It leads to coefficients swinging wildly, inverted signs and unreliable p-values. It doesn't affect the predictions but can make the interpretation of results difficult and unreliable.

2. Explain the Anscombe's quartet in detail. (3 marks)

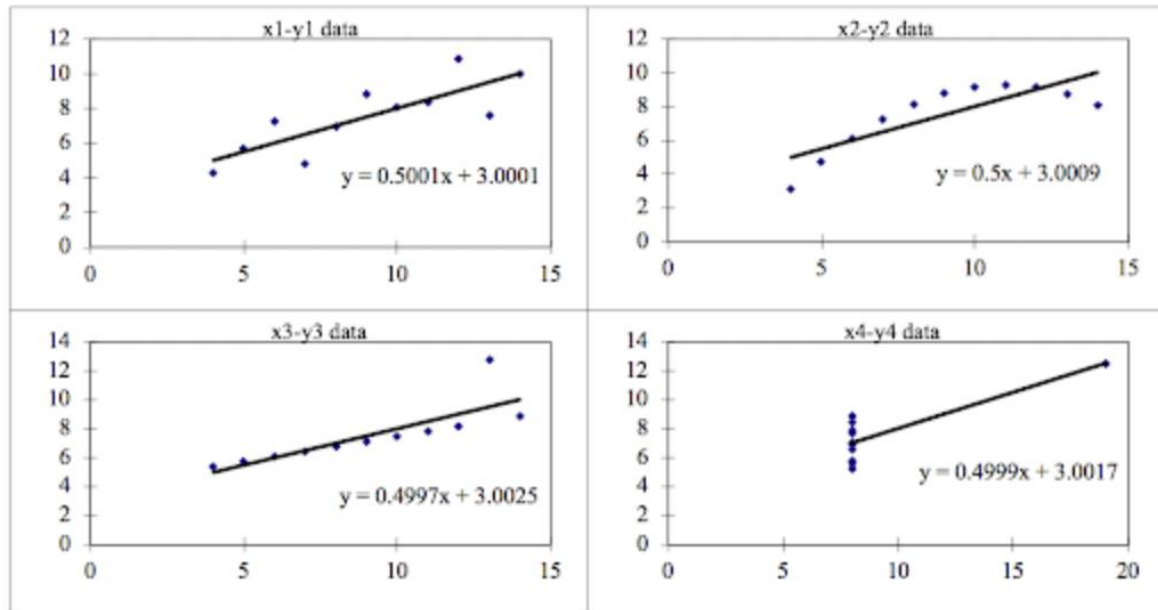
Anscombe's quartet was constructed in 1973 by statistician Francis Anscombe to illustrate the importance of plotting data before it is analysed and used to build the models.

It is a group of four datasets that are nearly identical from the point of view of simple descriptive statistics, but there are slight differences that fool the regression model once each dataset is actually plotted. The datasets have very different distributions so they look completely different from one another when the data is visualized on scatter plots.

The plots are defined as follows:

Anscombe's Data											
Observation	x1	y1		x2	y2		x3	y3		x4	y4
1	10	8.04		10	9.14		10	7.46		8	6.58
2	8	6.95		8	8.14		8	6.77		8	5.76
3	13	7.58		13	8.74		13	12.74		8	7.71
4	9	8.81		9	8.77		9	7.11		8	8.84
5	11	8.33		11	9.26		11	7.81		8	8.47
6	14	9.96		14	8.1		14	8.84		8	7.04
7	6	7.24		6	6.13		6	6.08		8	5.25
8	4	4.26		4	3.1		4	5.39		19	12.5
9	12	10.84		12	9.13		12	8.15		8	5.56
10	7	4.82		7	7.26		7	6.42		8	7.91
11	5	5.68		5	4.74		5	5.73		8	6.89
				Summary Statistics							
N	11	11		11	11		11	11		11	11
mean	9.00	7.50		9.00	7.500909		9.00	7.50		9.00	7.50
SD	3.16	1.94		3.16	1.94		3.16	1.94		3.16	1.94
r	0.82			0.82			0.82			0.82	

As can be seen from above, their descriptive statistics seem pretty similar, almost equal in some cases. But when they are actually plotted, each data set generates a different kind of plot and not all of them can fit on a straight line as shown below:



The essence of Anscombe's quartet revolves around the idea that data features must be plotted to see the distribution of the samples that can help in identification of the various anomalies present in the data (outliers, diversity of the data, linear separability of the data, etc.). Moreover, the linear regression can only be considered a fit for the data with linear relationships and is incapable of handling any other kind of data set.

3. What is Pearson's R? (3 marks)

Pearson's R, also known as **Pearson's correlation coefficient**, measures the strength and direction of a linear relationship or correlation between two continuous variables.

The correlation ranges from **1 to -1** where:

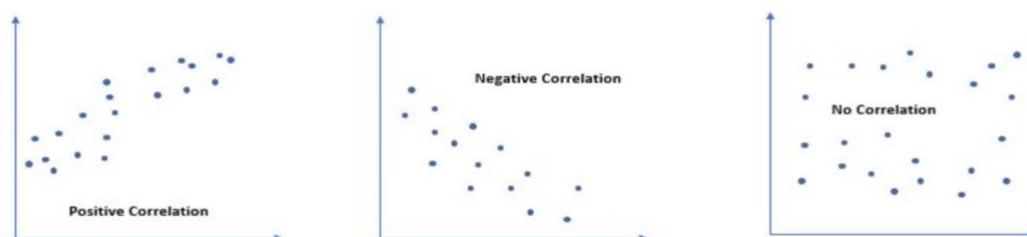
1 = perfect positive correlation

-1 = perfect negative correlation

0 = no correlation or linear relationship

Positive correlation here means that if a variable increases, the variable related to it also increases in some ratio and vice versa.

On the other hand, **negative correlation** means that if a variable increases, the variable related to it decreases in some ratio and vice versa.



Assumptions for Pearson correlation:

- Data must be from random or representative samples.
- Both variables should be continuous and should follow a normal distribution.
- Homoscedasticity is crucial.
- Extreme outliers impact the Pearson's correlation coefficient.

4. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling? (3 marks)

Scaling, also called **feature scaling**, helps in improving accuracy and efficiency of predictive models. It is used to transform the values of numeric variables in a dataset to a similar measurable scale.

Feature scaling is needed when dealing with datasets containing features/variables that have different ranges, units of measurement, or orders of magnitude. In such cases, the variation in feature values can lead to biased model performance or difficulties during the learning process. The purpose is to ensure that all features contribute equally to the model and to avoid the domination of features with larger values.

When to perform feature scaling?

- When working with gradient descent based algorithms like regression or neural networks
- When working with distance based algorithms like KNN, K-means clustering or SVM

Normalized Scaling

Normalized scaling is a method in which values are shifted and rescaled so that they end up ranging between 0 and 1. It is also known as Min-Max scaling.

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

X = the current value of variable column

Xmax = maximum value of variable

Xmin = minimum value of variable

Standardized Scaling

Standardized Scaling is a method where the values are centered around the mean with a unit standard deviation. This means that the mean of the attribute becomes zero, and the resultant distribution has a unit standard deviation.

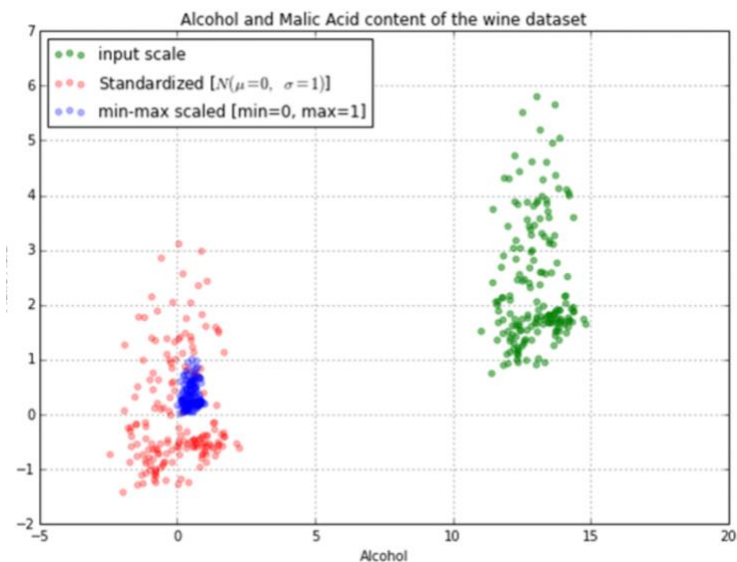
$$X' = \frac{X - \mu}{\sigma}$$

X = the current value of variable column

myu (u) = mean of the variable

sigma (s) = standard deviation of the variable

Normalized Scaling	Standardized Scaling
Rescales values to a range between 0 and 1	Centers data around the mean and scales to a standard deviation of 1
Useful when the distribution of the data is unknown or not Gaussian	Useful when the distribution of the data is Gaussian or unknown
Sensitive to outliers	Less sensitive to outliers
Retains the shape of the original distribution	Changes the shape of the original distribution
May not preserve the relationships between the data points	Preserves the relationships between the data points
Equation: $(x - \min)/(\max - \min)$	Equation: $(x - \text{mean})/\text{standard deviation}$



5. You might have observed that sometimes the value of VIF is infinite. Why does this happen? (3 marks)

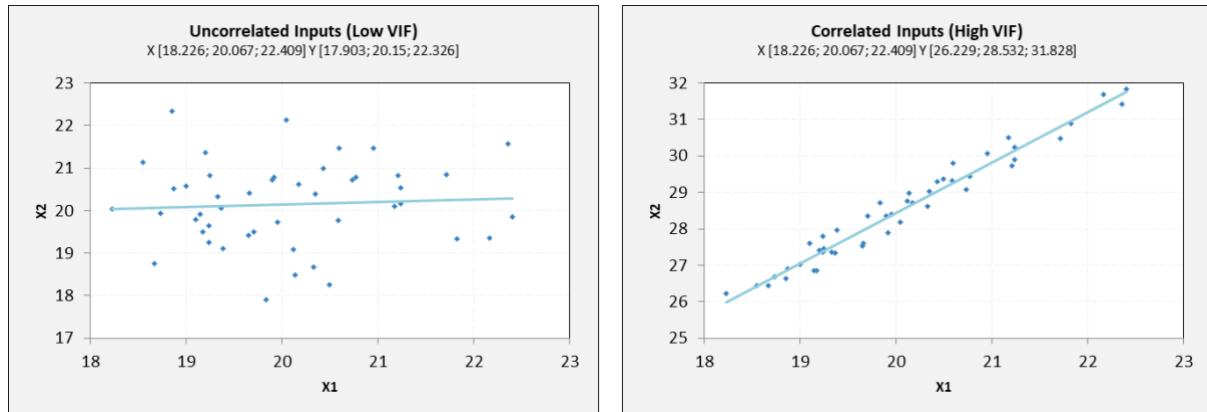
VIF, called **Variance Inflation Factor**, is a measure of the amount of multicollinearity in a set of multiple linear regression variables. It is one of the major statistics used to detect multicollinearity in a model.

VIF can be calculated as follows:

$$VIF = \frac{1}{1 - R_j^2}$$

As you can see, the more the chance of R2 score being closer to one, the higher the chance of VIF being infinite.

If there is perfect correlation between some variables, which can further lead to overfitting of model and highly inflated R2 score, VIF is going to be infinite.



6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression. (3 marks)

Q-Q plots, **also called Quantile-Quantile plots**, are used to plot the quantiles of a sample distribution against quantiles of a theoretical distribution. The first quantile is that of the variable you are testing the hypothesis for and the second one is the actual distribution you are testing it against.

Probability Distributions

In probability distributions, data is represented using plots where the x-axis represents the possible values of the sample and the y-axis represents the probability of occurrence.

There are various probability distribution types like Gaussian or Normal Distribution, Uniform distribution, Exponential distribution, Binomial distribution, etc.

Using Q-Q plots helps in determining if a dataset follows any particular type of probability distribution like normal, uniform, exponential. The power of Q-Q plots lies in their ability to summarize any distribution visually.

Q-Q plot can also be used to test distribution amongst 2 different datasets. For example, if dataset 1, the age variable has 200 records and dataset 2, the age variable has 20 records, it is possible to compare the distributions of these datasets to see if they are indeed the same.

Importance of Q-Q plot

It helps to determine if two data sets come from populations with a common distribution. This helps in a scenario of linear regression when we have training and test data set received separately and then we can confirm using Q-Q plot that both the data sets are from populations with same distributions. It can be used with sample sizes also. Many distributional aspects like shifts in location, shifts in scale, changes in symmetry, and the presence of outliers can all be detected from this plot. It is used to check if two datasets come from populations with a common distribution.